Beyond Exact Copies: Detecting Edited Copies of AI Agent Skills

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Abstract

Developers reuse agent skills, the `SKILL.md` instruction files that coding agents load on demand, by copying them between repositories and often editing them on the way. Without a package manager, nothing records where an edited copy came from. Existing studies link copies through registry names, fork relations, or identical versions, and similarity-based clone detection cannot tell an edited copy from a skill generated by the same template. We label 400 real skill pairs from the GitSkills dataset as derived variants, template siblings, or unrelated (Cohen's $\kappa = 0.95$) and train a classifier on lexical, boilerplate-aware, metadata, code, structural, and repository features, with boilerplate lines counted over all 1.59 million in-scope skills of the dataset. It identifies derived variants with an F1 of 0.94 (95% CI 0.89–0.97), against 0.74 for a SkillReuse-style baseline, and labels none of the 159 highly similar non-derived pairs as derived. Applied to 9.0 million candidate pairs, it predicts 4.1 million derivation links; a blinded manual check of 80 of them confirms 79 (precision 0.99, 95% CI 0.93–1.00). Counting reuse by derivation family instead of by exact file hash raises the Gini coefficient of reuse from 0.49 to 0.61–0.63 and the share of files in the top 1% of units from 27% to 40–42%, so reuse is markedly more concentrated than exact copies suggest.

1. Introduction

Agent-based assistants, such as Claude Code, have become part of the everyday software development workflow: they edit files, run commands, and review code changes on the developer's behalf. To make the agent perform a task according to a team's procedure, developers define this procedure as an *agent skill*. A skill is a directory that contains a `SKILL.md` file, which starts with YAML front matter holding a `name` and a `description`, followed by instructions written in Markdown; the directory may also contain scripts and reference documents [3, 11]. At first, the agent reads only the descriptions of the available skills, and it loads the full instructions of a skill only when the task matches its description. Anthropic introduced this format as an open standard in October 2025, and nine months later millions of `SKILL.md` files were stored in public GitHub repositories [11].

Libraries included in the application via npm or PyPI are mentioned in the manifest, fixed at a specific version, and installed with the help of the package manager. Skills lack such an installation mechanism. The developers copy a directory with the skill to their codebase with any modifications made on the way, without leaving any signs of the origin of the skill. The GitSkills dataset demonstrates how often this practice occurs: of 3,797,117 `SKILL.md` files collected from 282,200 repositories, 50.5% are exact binary copies of another file in the dataset [11]. Copies that were edited after copying are not included in this percentage, because any edit changes the content hash.

The following definitions are used throughout the paper. An *exact copy* has the same content hash as another file. A *derived variant* is a skill created by copying another skill and modifying it through renaming, trimming, extending, rephrasing, or translation; two skills copied from the same source are derived variants of each other. A *template sibling* has only a generic structure in common with another skill, such as section names, general instructions, or the style of a catalog, while the rest of its instructions were written independently. A group of skills connected by derivation forms a *skill family*, and the derivation links within a family form its *lineage*. We call reuse *hidden* when two skills are derived from one another but share neither a content hash nor a front-matter name, so that neither exact matching nor name matching can connect them. These notions are borrowed from clone detection in software engineering, where clones range from exact copies (Type-1) to copies with renamed identifiers (Type-2) and copies with added, removed, or changed statements (Type-3) [16].

Lineage matters because an agent executes a skill rather than merely displaying it. A person who reads an outdated README file may notice the problem and work around it, but an agent follows the outdated instructions [12]. Skills can also be dangerous: security audits have found both vulnerable and malicious skills among publicly available ones [14], and in January 2026 a single campaign added more than 1,200 malicious skills to the OpenClaw skill marketplace [20]. Once a flawed skill has been copied and modified, its descendants become hard to find.

To address these challenges, we investigate three research questions:

RQ1. *How accurately does our approach recognise derived skill variants, compared with exact hashing, similarity thresholds, and SkillReuse?*

RQ2. *Which signals contribute most to separating derived skill variants from template siblings and unrelated skills?*

RQ3. *How much skill reuse is hidden from exact and name matching, and how does counting it change the picture of reuse?*

2. Related Work

2.1. Reuse and adaptation of agent skills

Gao et al. [12] presented an early analysis of skills as engineered artifacts. They collected 18,463 skills from the skills.sh registry and 23,199 skills from 5,876 GitHub repositories. In order to connect a repository skill with a registry skill, they first required an exact match of the YAML `name`, then used longest-common-subsequence similarity to choose between candidates with the same name, and filtered out all the pairs with less than 0.1 similarity. This resulted in 3,709 reuse links. The results showed that reuse is often near-exact: 53% of the reused skills were never modified after being adopted. Among reused skills that were never updated locally, 40.2% of skills were taken from a registry whose source later changed, but those

copies were not updated. However, the approach can only detect copies with known source published on skills.sh and unmodified YAML name. The authors note that renamed and modified skills are not detected, and their approach presupposes that reuse goes from registry to repositories. In their dataset 76% of the collected repository skills are not linked, and some of them likely represent reuse that was not detected.

Wu et al. [19] studied how downstream developers adapt published skills by studying public forks of six popular skill repositories. Based on 1,126 adaptation cases, they offered a taxonomy with 46 adaptation patterns belonging to 13 families, and they report that 18.6% of the adaptations add security-sensitive content to the instruction text. They also call for tools that detect inconsistencies in modifications. Since their findings are based on GitHub fork relations, they only apply to copies created via forks of those six repositories; when a skill is pasted into an unrelated repository, no fork relation is left.

2.2. Reconstructing how skills spread

Seddik [18] reconstructs how skills spread in the same GitSkills release that we use. The study clones the default branch of each repository in the dataset, collects the git history of every `SKILL.md` file, and builds a dated, directed copy network of 2,193,119 skill adoptions. A few repositories turn out to be the source of almost all copies, and GitHub stars do not reveal them. Copies also seldom stay in sync with their sources: only 11.1% of the changes to a group of identical copies reach every copy in the group, and only 20.9% of copies adopt a later edit of their source. The network is built from exact content. Every unique content counts as one skill, and the versions of a skill are connected through the edits recorded by git. As we interpret the procedure, copies edited before their first commit share no version with their source and are therefore not connected to it; the same holds for renamed and translated copies. The study thus reveals from whom an exact copy was obtained and whether fixes propagate, but it leaves edited copies untouched.

2.3. Clone detection for agent skills

The skill reuse detector closest to ours is SkillReuse [20]. It represents the YAML metadata, prose, and code of a skill separately, and uses the similarities of these channels, together with a plain TF-IDF similarity and structural features, as inputs to a logistic regression that reaches an F1 score of 0.939 on SkillReuse-Bench, a collection of 300 labelled pairs. Applied to 137,470 skills crawled from the SkillsMP marketplace, it places 66.8% of skills in at least one clone pair, mostly across authors. The authors state explicitly what clone detection cannot show: structural similarity does not mean derivation, so detected pairs may include convergent development, and a clone relation does not say which skill is older. Two properties of the benchmark matter for us. First, half of its 150 positive pairs are synthetic mutations rather than real modifications. Second, its hard negatives are only same-category pairs whose similarity lies between 0.15 and 0.45, so highly similar pairs written independently, such as two skills generated from one template, are not included. Families formed as connected components of the clone graph also chain together: the largest one contains 61,651 skills. Near-duplicate detection has reached developer tools as well: drskill warns when two installed skills exceed a MinHash similarity threshold [1], and skillvariants groups near-clones of a skill found through GitHub code search [2]. Like SkillReuse, both tools find similar skills; neither decides whether one skill was copied from another.

2.4. Code clones and reuse on GitHub

The detection of code clones is a mature field. Roy et al. [16] characterise the standard clone types and evaluate detection techniques, and SourcererCC scales token-based detection of near-miss clones to very large code corpora [17]. Lopes et al. [15] showed that file-level duplication is pervasive on GitHub. Kim et al. [13] introduced clone genealogies, which track groups of clones across versions and separate consistent changes, made to all members of a group, from inconsistent ones. These methods rely on lexical and syntactic evidence, which is strong for source code. Skills, however, are mostly natural language and are often produced from common templates, so surface similarity is a weaker signal of shared origin.

2.5. Summary

Table 1 summarises the approaches. Each earlier approach either needs an exact record that survives copying or measures similarity without deciding origin; ours decides origin from content and validates the decision against human labels.

3. Problem Statement

A developer adopts an agent skill by copying the skill's folder into their own repository, possibly modifying its instructions along the way. No registry entry, manifest, or version pin records this adoption, and once a copy has been modified, its origin becomes difficult to trace. Three parties pay the price. Adopters are not told when the source of a skill they adopted is fixed, and because agents execute skills, a stale copy keeps following outdated or unsafe instructions. Skill authors do not know how far their skills have spread or what has been changed in them. Security reviewers who identify a harmful skill cannot find its edited copies.

Existing studies follow the reuse trail through a record that survives copying unchanged: the name of a skill published on a registry [12], a fork relation [19], or an exact version shared with the source together with the edit history around it [18]. In each case, the trail breaks when the copy is changed before being committed, renamed, or translated, since nothing exact is left to compare. Clone detection for skills [20] does not need any of these records, but it measures similarity rather than origin. It can treat two skills created from the same template as clones, and its benchmark contains no highly similar pairs that were written independently.

The problem therefore has two parts. First, to our knowledge, no existing method can distinguish an edited copy from a template sibling, so the reuse that exact matching misses cannot be told apart from shared scaffolding. Second, as a consequence, it is unknown how much reuse is hidden from exact matching. The problem has practical importance: when a reviewer discovers a harmful skill, the copy network can list its exact copies [18], but not the edited ones, which may still contain the harmful instructions.

Table 1: Summary of related work on tracing the reuse of agent skills.

Study	Data	How reuse is traced	Main finding	What it cannot do
Gao et al. [12]	18,463 registry and 23,199 GitHub skills	same YAML name, then text similarity	3,709 reuse links; 40.2% of unchanged copies miss later source changes	link renamed or heavily edited copies
Wu et al. [19]	forks of six popular skill repositories	GitHub fork relation	46 adaptation patterns; 18.6% add security-sensitive content	see copies pasted into unrelated repositories
Seddik [18]	GitSkills release, full git histories	identical versions joined by git edit history	2,193,119 adoptions; few repositories supply most copies	link copies edited before their first commit, renamed or translated
Zhu et al. [20]	137,470 marketplace skills; 300 labelled pairs	multi-channel similarity, logistic regression	F1 0.939; 66.8% of skills in a clone pair	separate copying from shared templates
This work	GitSkills release (1.59M skills); 400 + 120 labelled pairs	content classifier with boilerplate-aware features	F1 0.94; 0.99 precision of predicted links; family-level concentration	order derived pairs in time (Phase 4)

4. Proposed Solution

We propose to recognise derived variants from the content of skills alone, to distinguish them from template siblings, and to measure the reuse that exact matching misses. Our approach has two steps. The first step is a classifier that decides, for a given pair of skills, whether one is a derived variant of the other, a template sibling of it, or unrelated to it. Separating these three classes is the core difficulty. Skills are typically short, mostly written in English, and frequently scaffolded from the same templates, so two independently written skills can be very similar, while a reworded or translated copy can share little text with its source. No fixed similarity threshold separates these cases, and training a classifier on labels produced by such a threshold would only teach it the threshold back. We therefore built our ground truth by hand: team members independently labelled candidate pairs from GitHub according to a written rubric, and we measured their agreement. Because the sample spans all similarity levels, it deliberately includes the highly similar template siblings that SkillReuse-Bench leaves out. Besides lexical similarity, the classifier uses signals aimed at the template problem, including a similarity computed only over the shared lines that do not recur across many unrelated skills, together with front-matter, structural, and repository evidence. We compare it with exact hashing, MinHash-based and TF-IDF similarity, and a re-implementation of SkillReuse [20].

The second step applies the classifier to all candidate pairs of the full GitSkills release [9], groups skills into families, and measures hidden reuse: how many derived pairs share neither a content hash nor a name, how many of them cross owners, and how counting them changes the picture of how concentrated reuse is. Because these estimates rest on predictions, we check a random sample of predicted links by hand and report every estimate together with the measured precision.

This design avoids the limitations described above. It needs no registry entry, fork record, or shared version, which matters in practice: GitHub code search indexes a fork only when it has more stars than its parent repository, so only three repositories in GitSkills are forks [10]. It separates derivation from template similarity and validates this separation on real pairs rather than synthetic mutations. It forms families with community detection, so that a few bridging pairs cannot merge unrelated skills into one giant family. Finally, it complements the copy network of Seddik [18]: that network covers exact copies with their full history, while our approach targets the edited copies it cannot link. Both identify file contents by their git blob hash, so in Phase 4 we will match our derived links against that network directly.

Table 2: GitSkills fields used in this study.

Field(s)	Use
`file_sha`, `repo_full_name`, `path`, `location_class`	occurrences, exact-copy counts
`filename`, `dedup_primary`, `frontmatter_valid`, `body_chars`	scope filtering
`content`, `name`, `description`	skill text (representatives)
`first_commit_at`	history coverage
`artifact_siblings.entry_name`	bundled-file similarity
owner (prefix of `repo_full_name`)	same-owner and cross-owner reuse

5. Dataset Extraction

Source. The course provides GitSkills as a single SQLite file, `agent_skills_release.db` [9], with four tables: `artifacts`, with one row per file occurrence; `repos`, with repository metadata; `artifact_siblings`, with the files bundled with each skill; and `mining_runs`, with collection provenance [11]. We use it at two levels. The *full release* (3,797,117 file occurrences of 1,877,981 distinct contents in 282,200 repositories owned by 195,841 accounts) is the population for every analysis. The *official sample* that the dataset authors cut from the release [10], the 13,000 distinct contents with the lowest content hashes together with every row the release holds about them, is where we drew the pairs to label. Content hashes are effectively uniform, so the sample is a random sample of distinct skills; it has the same schema as the release, and every labelled pair also occurs in the release.

Fields. Table 2 lists the fields we extract.

Scope. The filename search behind GitSkills also matches lowercase `skill.md` files and similarly named files that predate the

Table 3: Scope filters applied to the official sample and to the full release.

Filter	Sample	Full release
All distinct contents	13,000	1,877,981
Exact basename `SKILL.md`	12,763	1,842,600
Valid YAML front matter	11,197	1,610,957
Body ≥ 200 characters (dataset field)	11,010	1,586,606
Body ≥ 200 characters after our own parsing	11,009	1,586,429

format, so we keep only representatives whose exact basename is `SKILL.md`. We also require valid front matter, because the name and description are inputs to our classifier, and a body of at least 200 characters, which removes stubs and symbolic-link skills whose content is only a link target. Table 3 shows the effect of each filter on both levels; the share of contents that pass is the same (85%), as expected for a uniform sample.

The working set of the full release contains 1,586,429 skills, which occur in 3,341,056 files in 258,245 repositories owned by 181,349 accounts. Commit history is available for 389,594 of these skills, 577,916 bundle at least one other file, and 158,210 front-matter names are carried by two or more distinct contents. The working set of the sample contains 11,009 skills.

Decomposition. We split each representative file into its normalised name, description, prose (the body without fenced code), code blocks, and headings, and add its repository context: the owners of all its copies, the name of its skill folder, and the names of its bundled files. This step writes a SQLite database with two tables, `skills` and `occurrences` (`extracted.db` for the sample, `full_skills.db` for the full release), which is the input of the following steps.

6. Study Design

Figure 1 gives an overview of our approach. It has five stages: the first two are automatic, the third is our manual analysis of the dataset, and the last two build the classifier and apply it.

Changes since Phase 2. Phase 3 extends the study in six ways, each answering a limitation of Phase 2. (i) The gold set doubles from 200 to 400 labelled pairs. (ii) Every analysis now runs on the full GitSkills release instead of the official sample. (iii) A blinded manual check measures the precision of the predicted links, so RQ3 is reported conditional on measured precision. (iv) A provenance check tests the labels against evidence that does not come from text similarity (Section 7.6). (v) The pipeline is made reproducible: package versions are pinned, the candidate pairs and the labelled sample are stored by content hash, threshold tuning no longer depends on floating-point details of a machine, fenced code is parsed as in CommonMark, and the labelling viewer is published with complete, untruncated skills. (vi) We report confidence intervals and significance tests, and we tried multilingual sentence embeddings to recover translated copies; they did not help (Section 7.4).

Stage 1: Data preparation. We extract the working set described in Section 5 and split each skill into its name, description, prose, code blocks, and headings, together with its repository context.

Table 4: Similarity bands of the candidate pairs of the official sample and of the pairs drawn from them for labelling.

Band	Definition	Candidates	Sampled
B1	$J \geq 0.9$	426	60
B2	$0.7 \leq J < 0.9$	615	80
B3	$0.5 \leq J < 0.7$	1,278	80
B4	$0.3 \leq J < 0.5$	1,347	80
B5	same name, $J < 0.3$	1,889	50
B6	LSA neighbour, $J < 0.3$	1,901	30
B7	random pair (not a candidate)	–	20
B0	other LSH match, $J < 0.3$	628	0
	Total	8,084	400

Stage 2: Candidate pair generation. Comparing every pair of the 1,586,429 skills would take more than a trillion comparisons, so we score only pairs proposed by inexpensive blockers. MinHash locality-sensitive hashing [5] over word 5-gram shingles of the body (128 permutations, 37 bands of 3 rows) proposes pairs whose estimated Jaccard similarity is around 0.3 or more, and same-name blocking proposes distinct contents that carry the same normalised front-matter name, capped at 45 pairs per name. On the sample we also used nearest neighbours in a latent semantic analysis (LSA) space [8] to catch reworded copies; in the labelled pairs, the pairs found only by LSA (band B6) contain no derived pair, so we drop this blocker at full scale. Large catalogs of near-identical skills fill single MinHash buckets with thousands of members; we pair each member of a bucket with more than 50 members with its 5 lowest-id members instead of with all of them, which keeps the bucket connected while the number of pairs stays linear (21,990 buckets were capped). On the full release the blockers propose 9,022,502 candidate pairs (Table 4 gives the bands of the sample, from which the labelled pairs were drawn).

Stage 3: Manual analysis. We followed three steps, explore, analyse, and synthesise, and added a fourth in Phase 3, validate.

Explore. Prior to annotation, we examined candidate pairs from all bands using a two-column side-by-side diff view along with selected skills to find out why particular skills resemble each other. Much of the similarity in the middle bands came from shared boilerplate rather than copying. For example, we found a collection of machine-generated skills named `u02156-...` that differed only in several words, skills that wrapped Rube MCP (Composio) boilerplate for multiple different applications, and also a series of German legal practice skills that featured the same checklist-like lines. On the other hand, there were several pairs with the same name and topic (such as two refactoring skills) which were created independently. These observations led to our three-class labelling scheme.

Analyse. Our stratified sample consisted of 400 pairs from all bands of Table 4, including 20 random pairs that no blocker had proposed, which serve as easy negatives. To keep the sample diverse, the same skill appears in at most two candidate pairs and the same name in at most four pairs; the random pairs were drawn without the skill limit, so one skill also appears in a third pair (P0044, P0145 and the random pair P0096). We randomised the left-right order of

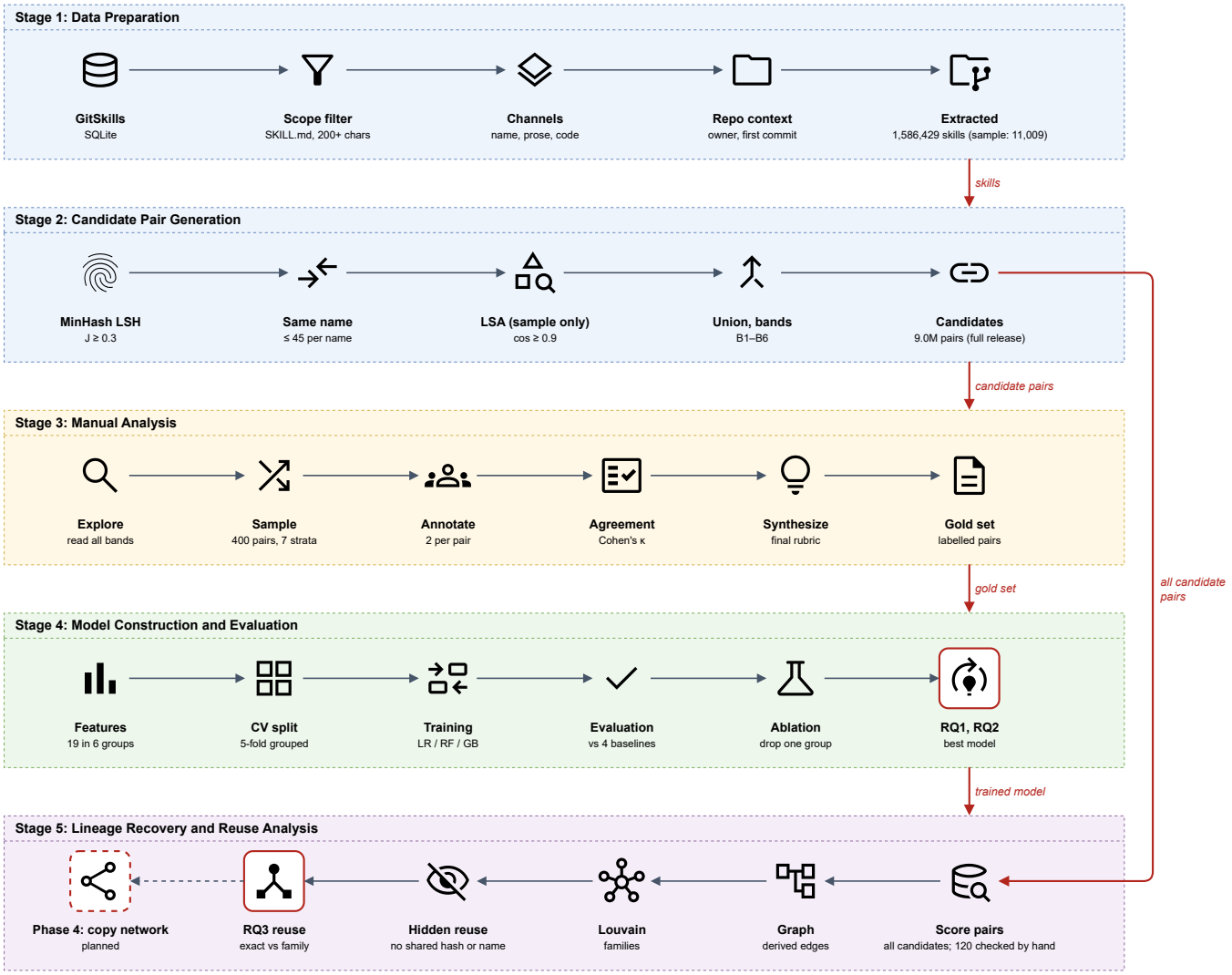


Figure 1: Approach overview. The shaded third stage is manual; outlined boxes answer our research questions, and the dashed box is planned for Phase 4. Stage 5 runs right to left, starting from the trained model and the full set of candidate pairs. The LSA blocker of Stage 2 is used on the official sample only.

each pair and did not show similarity bands or scores to the annotators, so that neither could bias their labels. Pair identifiers were assigned after shuffling, so the first 200 pairs, which we labelled in Phase 2, are a random subset of the sample; in Phase 3 we labelled the remaining 200, for 400 pairs in total. Two of the team members independently annotated each pair according to criteria adapted from those Gao et al. used to validate their reuse links [12]. A pair was derived when the skills share distinctive content that is unlikely to be produced independently (identical or slightly modified text, as well as task-specific steps, examples, commands, paths, and configuration values; renaming and translating do not change this). A pair was a template sibling if the only parts they shared were scaffolding text; otherwise the pair was unrelated, and the topic alone cannot be viewed as derivation. The annotators agreed on 388 of the 400 pairs, a Cohen's κ [7] of 0.95 over the three classes and 0.97 over the binarised data (i.e., derived vs. not derived). We resolved the 12 disagreements in discussion; all of them were borderline cases, such as a skill that reworks another one's steps without copying any line (P0121), or two skills that share a generated section schema but no content (P0185).[1]

Synthesise. From the notes we kept while labelling, we grouped the reasons why two skills look alike. Derived pairs arise from copy-and-edit, where a copy is trimmed or extended (P0000, P0046); from renamed copies (P0022, P0054, P0087); from translated copies (P0137); and from copies whose paths an installer rewrote for another agent tool (P0060, P0092). Template pairs arise from catalog generators that only fill slots (P0002, P0008), from integration catalogs that wrap many applications the same way (P0016, P0028),

[1]The labelling sheets and the gold set are part of our replication package: https://github.com/BhaveshMRA/RWPM.

from domain series written by one author (P0036, P0068), and from long project preambles shared by different skills (P0003, P0058). Unrelated pairs often share a name or a topic but no text (P0038, P0084). Resolving the disagreements fixed three rules: a standard format of the domain, such as the sections of an architecture decision record, does not make two skills template siblings (P0142); a shared generated skeleton does, even without shared content (P0185); and a skill that reuses another one's distinctive steps is derived even when no line is copied verbatim (P0121), as is an installer copy with rewritten paths (P0169). Of the 75 pairs that at least one of us marked as uncertain, 24 come from band B5, where two skills share a name but little text, and 31 join two skills from the same owner. The resulting gold set contains 400 pairs: 82 derived, 240 template, and 78 unrelated.

Validate. Labels on a sample stratified by similarity show how well the classifier separates the classes, but not how precise its predictions are on the population. After applying the classifier to the full release, we therefore drew 80 predicted derivation links at random, together with 40 random candidate pairs that the classifier predicted as not derived, shuffled them, and labelled all 120 pairs with the same rubric and viewer, again two of us per pair. Annotators did not know which pairs the classifier had predicted as derived. They agreed on every pair ($\kappa = 1.00$); we attribute this to the pairs themselves, which are dominated by near-identical copies and catalog templates, while the gold set deliberately oversamples the harder bands.

Stage 4: Model construction and evaluation. For each pair we compute 19 features in six groups (Table 5). Corpus-level statistics come from the full release: a normalised line (lower case, list markers removed, whitespace collapsed, at least 20 characters containing a letter) is *boilerplate* when it appears in skills carrying at least 5 different front-matter names, which gives 676,696 boilerplate lines (2,476 on the sample alone); the TF-IDF, LSA and normalisation models are fitted once on about 50,600 skills of the release (50,000 chosen at random plus every skill of the gold set), which keeps memory use manageable. We train logistic regression, random forest and gradient boosting classifiers with balanced class weights and evaluate them with five-fold stratified group cross-validation, so that no skill appears in both a training and a test fold (Section 7.2). Section 6.1 opens this stage in detail.

Stage 5: Lineage recovery and reuse analysis. We train the best classifier from Stage 4 on all gold pairs and apply it to every candidate pair of the full release. Pairs predicted as derived become the edges of a derivation graph, which we split into families with Louvain community detection [4]. Because a few wrong edges can join large groups of skills, we build the families twice: from all predicted links, and, as a conservative variant, only from links between skills with the same front-matter name, which renamed copies cannot enter. We then measure hidden reuse and compare reuse counted by exact hash with reuse counted by family (RQ3). As a check on the labels themselves, we also record, for every labelled pair, evidence that does not come from text similarity: explicit attribution to a source, shared non-generic bundled files, and whether both skills have a recorded first commit.

Table 5: Pair features used by our classifier.

Group	Features
Lexical	Jaccard and containment of word 5-gram shingles; TF-IDF cosine of the bodies; normalised edit similarity of the first 5,000 characters
Boilerplate-aware	Jaccard and containment of distinctive lines; Jaccard of boilerplate lines; share of shared lines that are boilerplate
Metadata	names equal; name similarity; TF-IDF cosine of the descriptions
Code	Jaccard of identifier tokens in code blocks; both skills contain code
Structure	similarity of heading sequences; length ratio; heading-count ratio
Repository	a shared owner; same skill-folder name; Jaccard of bundled file names

Reproducibility. Package versions are pinned (`requirements-full.txt`, with the complete environments of both machines in `requirements-lock.txt` and `requirements-mac.txt`). The candidate pairs of the sample are frozen in a file keyed by content hash, and the labelled sample is stored the same way, so rerunning the pipeline cannot change either. Threshold tuning rounds scores before choosing a threshold, so floating-point differences between machines no longer change baseline decisions. Fenced code blocks are parsed as in CommonMark, and the labelling viewer, which now shows complete skills, is published with the labels. The full-release run streams the database in chunks; run on the official sample, it reproduces the features of the sample pipeline exactly.

6.1. White-Box View of the Classifier

Figure 1 shows the classifier only as the Training box of Stage 4. Figure 2 opens it and follows one candidate pair from its two `SKILL.md` files to the decision.

(1) Input. A candidate pair (a, b) proposed by the blockers of Stage 2.

(2) Channel decomposition. Each file is split into its front-matter name n_s and description d_s, and its body into prose, code blocks and the sequence of headings H_s. From the body we derive the set of normalised lines L_s (lower case, list markers removed, whitespace collapsed, at least 20 characters containing a letter), the set of word 5-gram shingles S_s, and the set of identifiers K_s in code blocks (split at camel case and underscores). Repository context adds the owners O_s of all copies, the name of the skill's folder f_s, and the names of its bundled files F_s.

(3) Pair features. The two skills are compared channel by channel. Most features are a Jaccard similarity $J(X, Y) = |X \cap Y|/|X \cup Y|$, a containment $C(X, Y) = |X \cap Y|/\min(|X|, |Y|)$, a cosine of TF-IDF vectors, or an equality test. The boilerplate-aware group first removes the lines that belong to the boilerplate set B, so that $J(L_a \setminus B, L_b \setminus B)$ measures shared distinctive lines, and adds the share of shared lines that are boilerplate. Neither is computed per pair: B is counted once over all 1,586,429 skills of the release, the TF-IDF models are fitted once on about 50,600 of them, and both are reused for every pair.

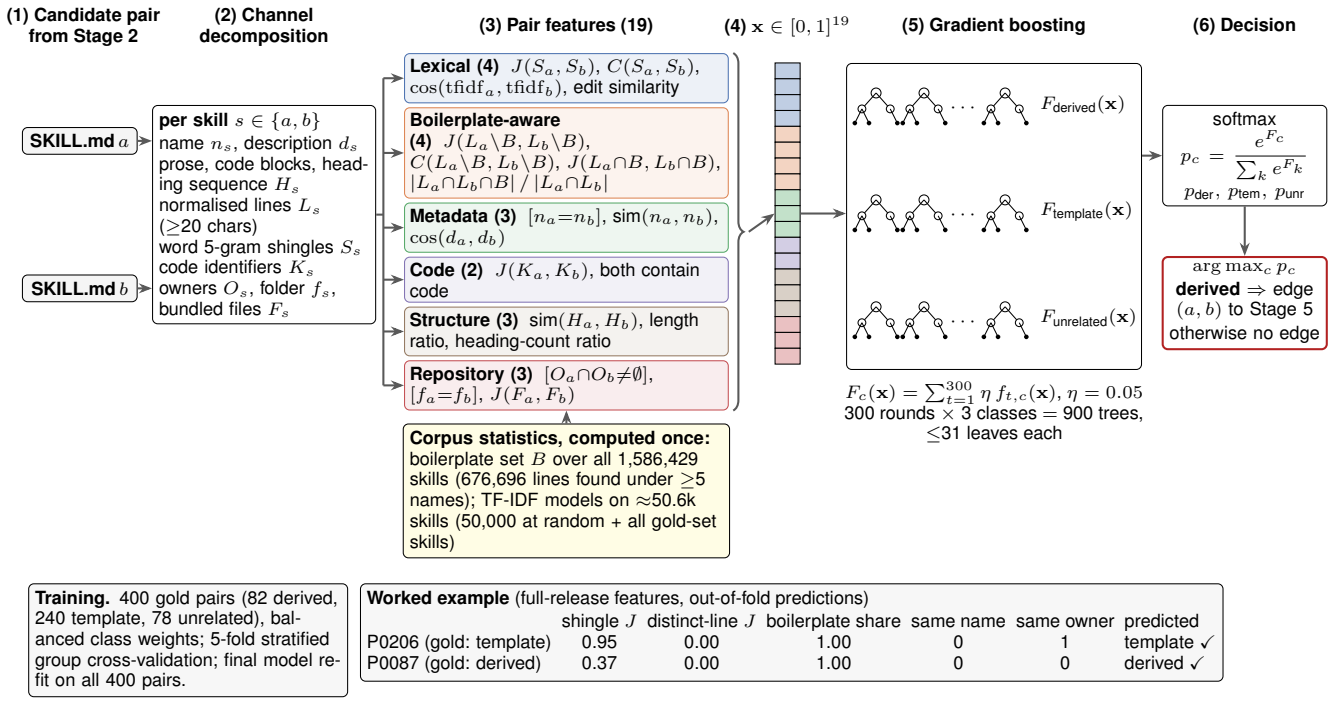


Figure 2: White-box view of the classifier (Stage 4 of Figure 1): how a candidate pair is decomposed into channels, turned into 19 features in six groups, and classified by gradient boosting; derived pairs become edges for Stage 5.

(4) Feature vector. The 19 values form a vector $\mathbf{x} \in [0, 1]^{19}$; every feature is a similarity, a ratio or a binary indicator.

(5) Gradient boosting. The classifier runs 300 boosting rounds. In each round it fits one regression tree per class (at most 31 leaves, at least 20 training pairs per leaf, features binned into at most 255 values) to the gradient of the multinomial loss, so the model consists of 900 trees. The score of class c is $F_c(\mathbf{x}) = \sum_{t=1}^{300} \eta\, f_{t,c}(\mathbf{x})$ with learning rate $\eta = 0.05$, and training weights each class inversely to its frequency, so that 82 derived pairs weigh as much as 240 template pairs.

(6) Decision. A softmax turns the three scores into probabilities, and the most probable class is the prediction. A pair predicted as derived becomes an edge of the derivation graph of Stage 5; template and unrelated pairs add no edge.

Worked example. The bottom of Figure 2 shows two labelled pairs. At full scale neither shares a distinctive line: the lines of both pairs occur under five or more names, so all their shared lines are boilerplate. The boilerplate-aware features therefore cannot separate them, and plain similarity would rank them the wrong way round, since the template pair P0206 shares far more text (shingle Jaccard 0.95) than the derived pair P0087 (0.37). The model still classifies both correctly by combining the other groups; for example, the two skills of P0206 belong to one owner, which in the linear view of Figure 3 is the strongest single indicator of a template.

7. Experiments

7.1. Research Questions

Three research questions assess our solution.

RQ1 (Effectiveness). How accurately does our approach recognise derived skill variants, compared with exact hashing, similarity thresholds, and SkillReuse? This question tests whether derivation can be recognised from content alone, and how our design performs on the pairs that matter most: highly similar pairs that are not derived. We report precision, recall, F1, and the Matthews correlation coefficient (MCC) [6] for the *derived* class, computed on pooled out-of-fold predictions, together with the standard deviation of F1 across folds and 95% bootstrap confidence intervals. We also report the false-positive rate on high-similarity pairs (non-derived pairs in bands B1–B3), precision and recall per band, and, for our approach, a three-class confusion matrix and macro-F1.

RQ2 (Contributing signals). Which signals contribute most to separating derived skill variants from template siblings and unrelated skills? We remove one feature group at a time, retrain the best model, and measure the change in F1 and in the high-similarity false-positive rate. We also train a model on lexical features alone.

RQ3 (Hidden reuse). How much skill reuse is hidden from exact and name matching, and how does counting it change the picture of reuse? This question measures what hash-, name-, and version-based approaches leave out. Among the labelled derived pairs, we measure the share whose front-matter names differ; since the two

skills of every pair also have different content hashes, these pairs are invisible to both exact and name matching. Among all derivations predicted by the best model, we measure the share with different names and the share that crosses owners. We then compare two units of reuse, exact hash and derivation family, by the number of units, the inflation ratio (distinct contents per family), the size of the largest family, the Gini coefficient of file occurrences per unit, and the share of units spanning two or more owners. We leave the comparison with the copy network of Seddik [18] to Phase 4.

7.2. Experimental Setup

RQ1 and RQ2 use the 400 gold pairs; RQ3 uses the predictions on the full release and the manually checked sample of them. We use five-fold stratified group cross-validation; groups are formed by connected components of the labelled-pair graph, so no skill contributes to both training and testing. We compare four baselines with our approach.

Normalised-text hash predicts *derived* only when the two bodies are identical after lower-casing and collapsing whitespace, which is what exact hashing sees once trivial formatting is ignored. *Shingle Jaccard (MinHash)* thresholds the exact Jaccard similarity of word 5-gram shingles, the quantity MinHash estimates. *TF-IDF cosine* thresholds the cosine similarity of TF-IDF vectors of the bodies. For both, the threshold that maximises F1 is chosen on the training folds only. *SkillReuse-style* is our re-implementation of the eight features of Zhu et al. [20]: a flat TF-IDF similarity, LSA cosine similarities for the YAML, prose, and code channels (set to zero when a channel is missing), their three pairwise products, and a structural similarity over eight document statistics, combined by a class-balanced logistic regression whose threshold is again tuned on the training folds. It is trained on our gold set, so it is compared with our approach on equal terms; it is not the authors' released implementation.

Our approach uses the 19 features of Table 5 with three classifiers from scikit-learn: logistic regression on standardised features, a random forest with 500 trees, and histogram-based gradient boosting, all with balanced class weights and no threshold tuning. The classifiers are trained on the three classes when the gold set has at least 15 template pairs, and on derived versus not derived otherwise. We report 95% bootstrap confidence intervals (2,000 resamples of the labelled pairs) and compare our best model with the strongest baseline using McNemar's exact test. On the sample we also tried adding multilingual sentence embeddings (`paraphrase-multilingual-MiniLM-L12-v2`) as a sixth feature group, aimed at translated and reworded copies (Section 7.4).

7.3. RQ1: Effectiveness

Table 6 compares all methods. The best configuration of our approach, gradient boosting, reaches a precision of 0.97, a recall of 0.90, and an F1 of 0.94 (95% CI 0.89–0.97; MCC 0.92); random forest is almost as good (F1 0.94), while logistic regression drops to 0.87 with full-corpus features, presumably because some feature distributions become very skewed. The strongest baseline, the SkillReuse-style detector, reaches an F1 of 0.74 (0.66-0.81). The difference in F1 is 0.13-0.27 (95% CI), and the two methods disagree in a way that is very unlikely by chance: ours is right on 36 pairs that the baseline gets wrong, the baseline on 3 (McNemar's test, p <0.001). Among the 159 highly similar pairs that annotators judged not derived, our approach labels none as derived, the SkillReuse-style detector 10%, and MinHash similarity 33%. Across the three classes our approach reaches a macro-F1 of 0.95 (Table 8).

The confusion matrix (Table 8) shows where the model succeeds and where it fails. It recognises template siblings almost perfectly: 238 of the 240 template pairs are classified correctly and none is called derived, so the confusion behind most baseline false positives does not occur. This is where the design pays off: a template pair shares mostly boilerplate and, typically, an owner, while its names and descriptions differ in the substituted slots. The errors lie elsewhere: eight derived pairs are missed (four called template, four unrelated), two unrelated pairs are called derived, and two template and two unrelated pairs are confused with each other, which affects no derivation link.

Of the 96 false positives of the three similarity baselines, 90 are template pairs, and of the 19 derived pairs that all three miss, our approach finds 13; examples are P0054, which renames the grill-me skill and rewrites its rules around one copied sentence, and P0169, an installer copy of the gsd-manager skill whose paths were rewritten. On the derived class our model makes ten errors (eight missed copies and two false alarms), in three groups. First, four copies share no text with their source: P0121 reworks another skill's steps, P0137 and P0268 are Chinese adaptations, and P0335 keeps only the description; with no shared text, the model calls them unrelated. Second, four copies of two very popular skills, Matt Pocock's improve-codebase-architecture (P0053, P0123, P0297) and Superpowers' verification-before-completion (P0348), three of them renamed, are called templates: at full scale their lines occur under five or more names and therefore count as boilerplate. Third, the two false positives, P0226 and P0270, are unrelated skills that share a name and a topic.

> **Answer to RQ1:** Our approach identifies derived variants with an F1 of 0.94 (0.89-0.97), significantly better than the strongest baseline (0.74), and labels no highly similar non-derived pair as derived. Its errors are copies without shared text, copies of very popular skills, and same-name skills on the same topic.

7.4. RQ2: Contributing Signals

Table 9 reports the ablation. Lexical similarity alone is far from enough: a lexical-only model drops to an F1 of 0.59 and labels a quarter of the highly similar non-derived pairs as derived. No single other group is indispensable; removing any one costs between 0.02 and 0.05 in F1, most for metadata (0.05). Figure 3 shows what the best model relies on. On held-out folds, shuffling the metadata features costs most (0.33 in F1), followed by lexical (0.21), repository (0.08) and code (0.07) features; boilerplate-aware features cost only 0.05. On the sample alone, boilerplate-aware features had been the most important group (0.39). With boilerplate counted over the full release, the lines of popular skills that are copied under many names become boilerplate too, so these features separate templates from copies less sharply and the model leans on names and descriptions instead. Adding multilingual sentence embeddings did not help: on the sample, F1 changed from 0.944 to 0.938 (McNemar's test,

Table 6: RQ1: detection of derived variants (pooled out-of-fold predictions; F1 ± standard deviation across folds). The last column is the share of non-derived pairs in bands B1–B3 that a method labels as derived (lower is better).

Method	P	R	F1	MCC	FP rate (hi-sim)
Normalised-text hash	0.71	0.06	0.11±0.07	0.17	0.01
Shingle Jaccard (MinHash), tuned threshold	0.46	0.54	0.49±0.13	0.35	0.33
TF-IDF cosine, tuned threshold	0.59	0.37	0.45±0.08	0.36	0.13
SkillReuse-style (re-implemented)	0.73	0.76	0.74±0.05	0.67	0.10
Ours (logistic regression)	0.86	0.88	0.87±0.05	0.83	0.06
Ours (random forest)	0.99	0.89	0.94±0.05	0.92	0.00
Ours (gradient boosting)	0.97	0.90	0.94±0.06	0.92	0.00

Table 7: RQ1 by similarity band: precision (P) and recall (R) of the strongest baseline and of our best model.

				Best baseline		Ours	
Band	Definition	Pairs	Derived	P	R	P	R
B1	$J \geq 0.9$	60	33	0.92	1.00	1.00	1.00
B2	$0.7 \leq J < 0.9$	80	21	0.65	0.81	1.00	1.00
B3	$0.5 \leq J < 0.7$	80	7	0.60	0.86	1.00	0.71
B4	$0.3 \leq J < 0.5$	80	13	0.86	0.46	1.00	0.85
B5	same name, $J < 0.3$	50	8	0.00	0.00	0.67	0.50
B6	LSA neighbour, $J < 0.3$	30	0	–	–	–	–
B7	random pair	20	0	0.00	–	–	–

Table 8: RQ1: confusion matrix of our best model (out-of-fold).

True ↓ / Predicted →	*derived*	*template*	*unrelated*
derived	74	4	4
template	0	238	2
unrelated	2	2	74

Table 9: RQ2: ablation of feature groups for our best model.

Configuration	F1	ΔF1	FP rate (hi-sim)
All features	0.94	–	0.00
without lexical	0.90	-0.04	0.03
without boilerplate	0.91	-0.03	0.01
without metadata	0.88	-0.05	0.01
without code	0.91	-0.03	0.01
without structure	0.92	-0.02	0.01
without repository	0.91	-0.03	0.01
lexical only	0.59	-0.34	0.25

$p = 1.0$), the embeddings fixed two pairs and broke three, and the translated copies they were meant to catch stayed undetected, because embedding similarity measures topic as much as origin.

> **Answer to RQ2:** Signals beyond lexical similarity are essential (F1 0.59 with lexical features alone). At full scale, names and descriptions contribute most and boilerplate-aware features least, because boilerplate counted over 1.59 million skills also absorbs the lines of popular copied skills.

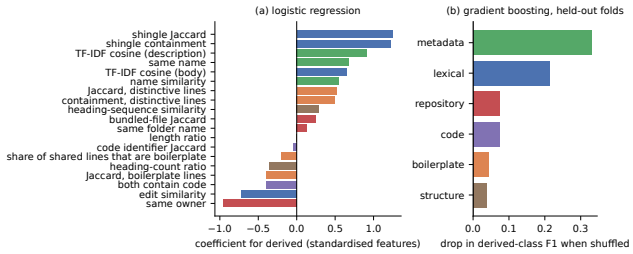


Figure 3: What the trained model relies on (complementing the white-box view of Figure 2): (a) logistic-regression coefficients for the derived class on standardised features; (b) drop in derived-class F1 of the best model on held-out folds when the features of one group are shuffled together.

7.5. RQ3: Hidden Reuse

Applied to the 9,022,502 candidate pairs of the full release, the best model predicts 4,058,198 derivation links. Of 80 of these links drawn at random, our manual check confirms 79, a precision of 0.99 (Table 10): all 60 sampled links between same-name skills and 19 of the 20 links between skills with different names were derived, and none of the 40 decoys was. Of the predicted links, 18.0% join skills with different names, which neither exact nor name matching can connect (24.4% of the labelled derived pairs); with the measured precision of such links (0.95, 95% CI 0.76-0.99), about one in six predicted derivations is hidden reuse. 92.4% of the predicted links cross owners.

Table 10: Manual check of predicted links on the full release (precision with 95% Wilson interval).

Links	Checked	Derived	Precision
All predicted links	80	79	0.99 [0.93, 1.00]
same name	60	60	1.00 [0.94, 1.00]
different names	20	19	0.95 [0.76, 0.99]
Decoys (predicted not derived)	40	0	–

Table 11: RQ3: reuse counted by exact hash and by derivation family on the full release.

Metric	Exact hash	Family all links	Family same name
Units	1,586,429	1,117,691	1,169,540
Gini of occurrences per unit	0.49	0.63	0.61
Occurrences in top 1% of units	27.0%	42.0%	40.1%
Units spanning ≥2 owners	13.4%	16.2%	16.0%
Occurrences in multi-owner units	52.3%	65.7%	64.5%

Table 11 compares reuse counted by exact hash with reuse counted by family, for families built from all links and from same-name links only.

Both variants tell the same story: counting families instead of hashes raises the Gini coefficient from 0.49 to 0.61–0.63 and the share of all files held by the top 1% of units from 27% to 40–42%. Popular skills exist in hundreds of edited versions that exact hashing counts as unrelated units: the frontend-design family contains 966 distinct contents with 4,564 files, and skill-creator, systematic-debugging, writing-plans and brainstorming follow. The two variants differ in their largest groups. With all links, the largest connected component has 4,564 skills; it and the next four join different skills of one ecosystem (the gstack commands, frontend design skills, the Composio integration catalog, skill-creator and related skills, and the spec-kit commands) through a small number of links between different skills. Such links are rare among all predictions, so the sample of 80 contains only one wrong link, but a few of them suffice to merge families. With same-name links only, the largest family has 966 skills, each of the ten largest families is a single skill, and the concentration figures barely change. The copy network of Seddik [18] finds that a few repositories supply almost all exact copies; our figures show, from the side of the copied skills, that reuse is also concentrated on a few skills, and more strongly once edited copies are counted.

> **Answer to RQ3:** About one in six predicted derivations joins skills with different names and is therefore hidden from exact and name matching; most derivations cross owners. Counted by family, reuse is markedly more concentrated than counted by exact hash (Gini 0.61–0.63 instead of 0.49), with or without the links between differently named skills. These figures are conditional on the predicted links, whose precision we measured at 0.99 (0.93-1.00).

Table 12: Provenance evidence per gold-set class.

Evidence	derived (n = 82)	template (n = 240)	unrelated (n = 78)
Explicit attribution to a source	8.5%	0.0%	0.0%
Names the other skill's repository or account	0.0%	0.0%	2.6%
Shared non-generic bundled file	22.0%	23.8%	0.0%
Both have a recorded first commit	4.9%	5.0%	6.4%

7.6. Validity of the Labels

Our labels judge content against a rubric; they do not observe copying. Table 12 checks them against evidence that does not come from text similarity. An explicit attribution to a source, such as "Credits: original skill by @blader" or "adapted from mattpocock/skills", appears in 8.5% of derived pairs and in none of the template or unrelated pairs; one attribution confirms P0137, a translated copy that our model misses. Such statements are rare, so they confirm labels rather than replace them. Shared bundled files do not separate the classes, because template catalogs ship the same scripts with every skill, and both skills of a pair have a recorded first commit in only about 5% of pairs, too few to order derived pairs in time. In the validation sample, annotators also found an edited copy that appends a reverse-shell section to an otherwise verbatim ClickHouse skill (V103, from a benchmark of malicious skills): exactly the kind of edited relative that exact matching cannot connect to its source.

8. Threats to Validity

Conclusion validity. Our conclusions rest on limited samples: the gold set has 400 pairs, only 82 of them derived, so one error moves recall by about a point, and the precision of links between differently named skills rests on 20 checked links (95% CI 0.76-0.99). We therefore report bootstrap confidence intervals and compare our model with the strongest baseline using McNemar's exact test instead of comparing point estimates. The measurement itself, human labels, is checked by double labelling: agreement is 0.95 on the stratified gold set and 1.00 on the validation sample, whose pairs are mostly clear-cut and therefore test the rubric less.

Internal validity. We chose the best of three classifiers by cross-validated F1, which slightly favours our approach; the random forest performs almost as well, so the conclusions do not hinge on that choice. Some features may capture correlates rather than causes of derivation: the same-owner feature can act as a shortcut for "template", because many catalogs belong to a single owner, and at full scale the boilerplate threshold of five names also covers the lines of popular skills copied under many names, which shifts the model towards metadata and accounts for four of its ten errors on the derived class. Families depend on a few links between different skills; we bound this effect with the same-name variant. At full scale we dropped the LSA blocker; in the labelled sample, the pairs it alone proposed contain no derived pair.

Construct validity. The labels come from four graduate students applying a written rubric, not from the authors of the skills. A derived label means that two skills share distinctive content that is unlikely to be written twice independently; it does not prove historical copying, and explicit attribution confirms only a small share of derived pairs (Section 7.6). Hidden reuse is measured by front-matter names, so a renamed copy counts as hidden even when

its folder name reveals the source. Our indicators are chosen for the task: F1 and MCC on the derived class, the false-positive rate on highly similar non-derived pairs, which measures the template problem directly, and the precision of predicted links; recall on the full release is observed only through the decoys of the validation sample.

External validity. We study one dataset: public GitHub repositories, default branches only, as indexed by GitHub code search, collected in July 2026, early in the adoption of the format [11]. Skills in registries and marketplaces such as skills.sh or SkillsMP, and in private repositories, may be reused differently. The data are real repositories, not synthetic mutations, and the 1.59 million skills come from 258,245 repositories of very different kinds; the labelled pairs come from a uniform sample of the release, although stratification oversamples similar pairs. Candidates come from same-name blocking and MinHash, so heavily reworded copies under a new name may never be scored, while classification errors can move the hidden-reuse share either way; the share is therefore an estimate for the predicted links, not a bound. Our SkillReuse baseline is reimplemented from the paper and trained on our labels; its numbers are not comparable to those of Zhu et al. [20].

9. Conclusion

Pairs of agent skills that our rubric labels as edited copies can be recognised from content alone and separated from template siblings: on 400 labelled pairs our classifier reaches an F1 of 0.94, against 0.74 for a SkillReuse-style detector, and it labels none of the highly similar template and unrelated pairs as derived. On the full GitSkills release, 79 of 80 of its predicted links are confirmed by hand. Counting these links shows that reuse is far more concentrated than exact copies suggest, and that about one in six predicted derivations is invisible to exact and name matching.

10. Future Work

In Phase 4 we will (i) scale the boilerplate threshold with the size of the corpus, so that the lines of popular copied skills stay distinctive, and limit links through bucket anchors; (ii) label more cross-language and heavily reworked copies, the main source of missed derivations; (iii) match our derived links against the copy network of Seddik [18] through the git blob hashes both datasets share, to measure how many of our links that network lacks and to order derived pairs in time; and (iv) release the classifier as a tool that lists the edited relatives of a given skill.

Data Availability

Our scripts, labelling sheets, gold set, validation labels and generated results are available at https://github.com/BhaveshMRA/RWPM/tree/phase3.

Acknowledgments

We used an AI assistant (Claude Opus 5.5, Anthropic) for brainstorming the research direction; for searching the literature, which led us to the Skill Constellations study [18]; for help in debugging the analysis scripts and assisting with the approach overview figure; for a review of labels and notes after the annotators had finished labelling, which changed some confidence values and notes but no labels; and for suggestions during the adjudication of disagreements, where the final decisions were ours. All labels were assigned by the authors, and the authors take full responsibility for the content of this paper.

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